ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

CIVIL TECHNOLOGY

THEORY PAPER

MARKS: 100 **TIME: 2 hours**

Topics: safety and equipment, concrete and materials, applied mechanics (beam reactions, centroids, stress and strain), construction, quantities and graphics. Suitable for Civil Services, Construction and Woodworking learners.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL formulae, substitutions and units in calculations. Round answers to TWO decimal places.
2. Make neat sketches where asked. A non-programmable calculator may be used.

QUESTION 1: MULTIPLE CHOICE

1.1 The ratio of the mass of water to the mass of cement in a concrete mix is called the …

A water–cement ratio. B slump.
C aggregate ratio. D yield.

1.2 A temporary mould that holds fresh concrete in shape until it has hardened is called …

A formwork (shuttering). B scaffolding.
C a screed. D a damp-proof course.

1.3 In a reinforced concrete beam resting on two supports, the main steel bars are placed near the …

A top of the beam. B bottom of the beam.
C centre of the beam. D sides only.

1.4 The test used on site to check the workability of fresh concrete is the …

A slump test. B cube crushing test.
C silt test. D bulking test.

1.5 The horizontal member of a roof truss that joins the feet of the rafters and is in tension is the …

A tie beam. B purlin.
C ridge board. D strut.

(5 × 2)
[10]

QUESTION 2: SAFETY AND EQUIPMENT

2.1 Give THREE safety requirements for erecting and using scaffolding. (3)

2.2 State the purpose of: (a) a dumpy level; (b) a poker vibrator. (4)

2.3 Explain what the Occupational Health and Safety Act means by a *competent person*. (2)

2.4 Give THREE safety precautions when working in a trench deeper than 1,5 m. (3)

2.5 Explain why the cutting disc of an angle grinder must be rated for at least the speed (r/min) of the grinder. (3)

[15]

QUESTION 3: CONCRETE AND MATERIALS

3.1 A floor slab 5 m long, 4 m wide and 150 mm thick is cast with a 1 : 2 : 4 mix (cement : sand : stone). The volume of dry materials is 1,5 times the volume of the wet concrete. 1 m³ of cement has a mass of 1 440 kg and cement is sold in 50 kg bags.

3.1.1 Calculate the volume of wet concrete and the volume of dry materials. (3)

3.1.2 Calculate the volume of cement, sand and stone needed. (3)

3.1.3 Calculate the number of bags of cement that must be bought. (3)

3.2 Describe, step by step, how a slump test is carried out. (5)

3.3 State TWO effects of adding too much water to a concrete mix. (2)

3.4 Name TWO admixtures for concrete and state the purpose of each. (4)

[20]

QUESTION 4: APPLIED MECHANICS

4.1 The simply supported beam AB in FIGURE 4.1 is 6 m long and carries three point loads. Ignore the mass of the beam. Calculate the reactions at A and B. (6)

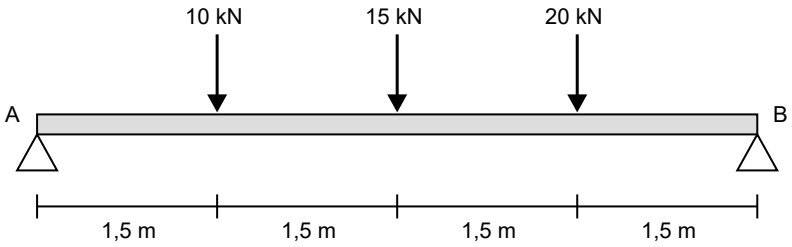


FIGURE 4.1

4.2 A T-section is made of a flange 120 mm wide and 20 mm thick on top of a web 20 mm wide and 100 mm high (total height 120 mm). Calculate the position of the centroid measured from the bottom of the web. (6)

4.3 A steel tie rod 2 m long with a diameter of 20 mm carries a tensile load of 50 kN. Young's modulus for steel is 200 GPa. Calculate: (a) the stress in the rod in MPa; (b) the strain; (c) the elongation of the rod in mm. (7)

4.4 Define: (a) stress; (b) strain; (c) Hooke's law. (6)

[25]

QUESTION 5: CONSTRUCTION

5.1 Give FOUR requirements of good formwork. (4)

5.2 Explain why the formwork under a concrete beam (the soffit) stays in place much longer than the side formwork. (2)

5.3 Name FOUR members of a king-post roof truss. (4)

5.4 Distinguish between a strip foundation and a raft foundation, and state when a raft foundation is used. (4)

5.5 Give TWO advantages of a cavity wall and state the purpose of wall ties. (3)

5.6 A staircase must rise 2,7 m from floor to floor. Each riser is 180 mm. Use the rule 2R + G = 600 mm to calculate: (a) the number of risers; (b) the going (G) of each step. (3)

[20]

QUESTION 6: QUANTITIES AND GRAPHICS

6.1 The floor of a room measures 4,2 m × 3,6 m inside. It will be covered with 400 mm × 400 mm floor tiles. Calculate the number of tiles needed, then add 10% for cutting and breakage. Round up to whole tiles. (5)

6.2 Name FIVE details that must appear in the title block of a building drawing. (5)

[10]

TOTAL: 100

END OF QUESTION PAPER